What is AR and VR?

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Introduction

In today's Modern era everyday lives is merged with Technology right from

- Cloud computing
- Artificial Intelligence
- Social media
- Machine Learning
- Big Data
- VR AR and many more

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Agenda

01 Introduction to VR/AR

02 Applications of VR AR

03 VR VS AR

04 Advantages and Disadvantages of VR/AR

05 Companies Using and Hiring VR/ AR

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What Is VR?

VR stands for Virtual Reality, and it is the use of computer technology to create a simulated environment that can be explored in 360 degrees

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Applications of VR

Education

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Applications of VR

Education

Healthcare

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Applications of VR

Education

Healthcare

Entertainment

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Applications of VR

Education

Automotive Industry

Healthcare

Entertainment

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Applications of VR

Education

Automotive Industry

Healthcare

Defence

Entertainment

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Applications of VR

Education

Automotive Industry

Healthcare

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Entertainment

Digital Marketing

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- In **education** conducting academic activities like field trips visit to museums and historical eras is now easy via vr to conduct virtuality with no change in common factors

- Towards healthcare analysis and research on medical terms via has made medicine practitioners enter the next level to avoid direct contact with harmful substances and hazardous risk factors

- Real-time experience of fictional characters or sci-fi movies animations and motions can be experienced by all with the use of vr

- In the field of entertainment prototyping cars help the automotive industry to avoid multiple designs and reduce resources to maximum via virtual designs via vr

- In terms of defense user vr helps a brave men to experience the battlefield environments in real time to avoid unconditional situations in reality

- Towards marketing vr promotes products where consumers can look and feel the product in real time via a 360 degree view which helps in better marketing

Beyond gaming and other entertainment cases, some business examples of virtual reality include:

- Architects are using VR to design homes — and let clients "walk through" before the foundation has ever been laid.
- Automobiles and other vehicles are increasingly being designed in VR.
- Firefighters, soldiers and other workers in hazardous environments are using VR to train without putting themselves at risk.

What Is AR?

AR: Augmented reality (AR) is an enhanced version of real-world entities that makes digital content appear to be a part of the physical world

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Applications of AR

AR Glasses

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Applications of AR

AR Glasses

Medical Imaging

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Applications of AR

AR Glasses

Medical Imaging

Entertainment

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Applications of AR

AR Glasses

Tourism

Medical Imaging

Entertainment

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Applications of AR

- AR Glasses
- Medical Imaging
- Entertainment
- Tourism
- Education

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Applications of AR

- AR Glasses
- Medical Imaging
- Entertainment
- Tourism
- Education
- Design and Modelling:

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What are some examples of augmented reality and virtual reality?

Augmented reality entails abundant — and growing — use cases. Here are some actual applications you can engage with today.

- Ikea Place is a mobile app that allows you to envision Ikea furniture in your own home, by overlaying a 3D representation of the piece atop a live video stream of your room.
- YouCam Makeup lets users virtually try on real-life cosmetics via a living selfie.
- Repair technicians can don a headset that walks them through the steps of fixing or maintaining a broken piece of equipment, diagramming exactly where each part goes and the order in which to do things.
- Various sports are relying on augmented reality to provide real-time statistics and improve physical training for athletes.

VR vs AR

Mode	VR	AR
 **Virtuality**	VR is completely virtual	AR uses a real-world entities enhancing digital content

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VR vs AR

Mode	VR	AR
Virtuality	VR is completely virtual	AR uses a real-world entities enhancing digital content
Control	VR users are controlled by the system	AR users can control their presence in real world

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Suggested: 5 Things You Should Know About Artificial Intelli...
Info

VR vs AR

Mode	VR	AR
Virtuality	VR is completely virtual	AR uses a real-world entities enhancing digital content
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Compatibility	VR requires compatible devices like a headset device	AR needs no specific device

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VR vs AR

Mode	VR	AR
Virtuality	VR is completely virtual	AR uses a real-world entities enhancing digital content
Control	VR users are controlled by the system	AR users can control their presence in real world
Compatibility	VR requires compatible devices like a headset device	AR needs no specific device
Environment	VR only enhances a fictional reality	AR enhances both the virtual and real world

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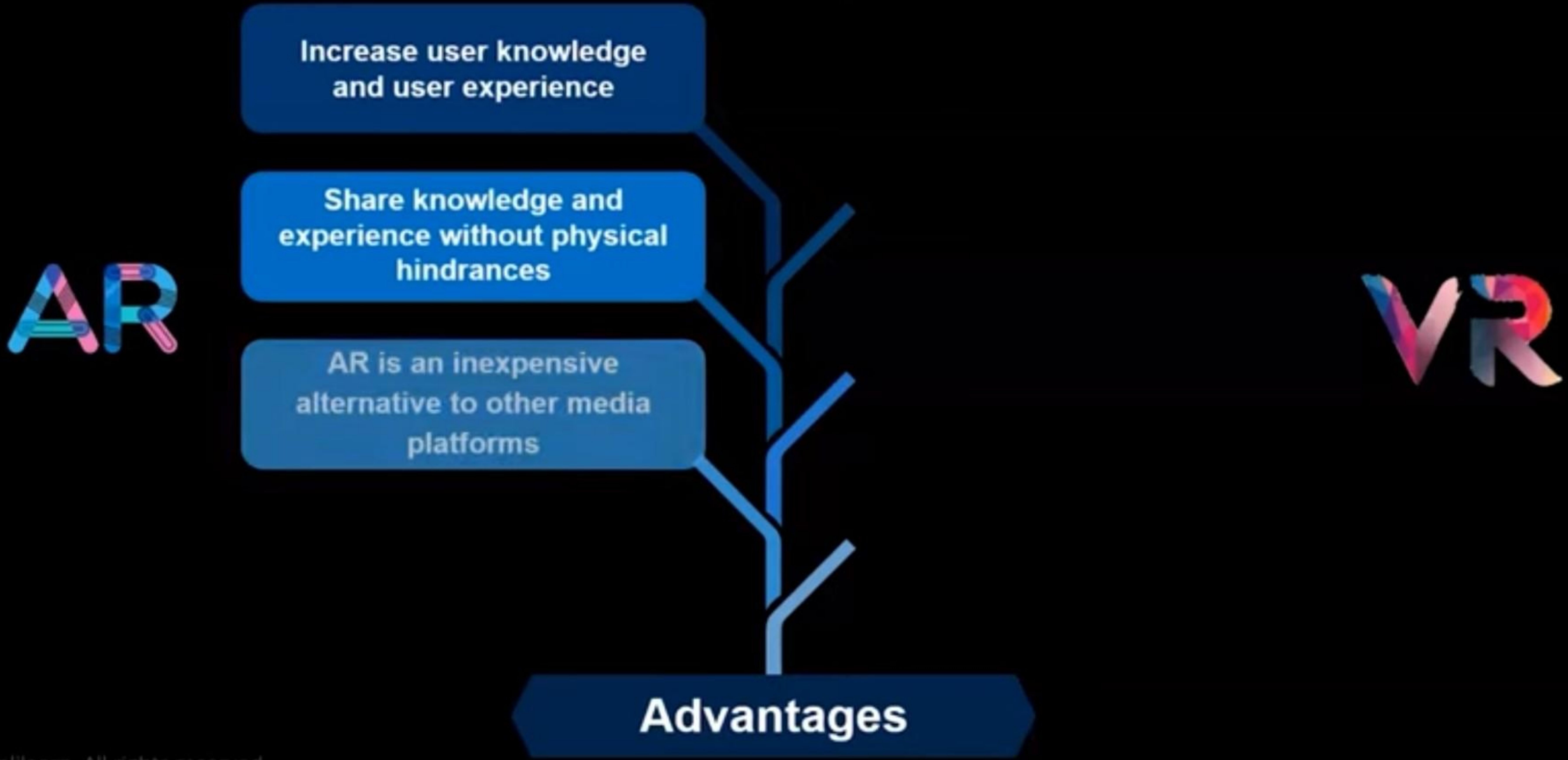
Advantages of AR and VR



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Advantages of AR and VR

AR

- Increase user knowledge and user experience
- Share knowledge and experience without physical hindrances
- AR is an inexpensive alternative to other media platforms

VR

- Virtual Reality makes the process more easy and comfortable
- With the use of VR, users can experiment with artificial environment
- Specifically designed environments to avoid misuse and Lack of Privacy

Advantages

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Disadvantages of AR and VR

AR

Quite expensive and less accessible

Can lead to use in inappropriate situations causing harm

AR technology is not equipped with security policies

Disadvantages

VR

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Disadvantages of AR and VR

Disadvantages

AR

- Quite expensive and less accessible
- Can lead to use in inappropriate situations causing harm
- AR technology is not equipped with security policies

VR

- Programmers sometimes find it difficult to interact with it
- Maximum use of VR makes addicted live in virtual world instead of real one
- VR take up a lot of space and require a lot of computing power

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Question: Are AR and VR Booming the Industry?

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Question: Are AR and VR Booming the Industry?

!NO!

In Addition to AR and VR there Exist other Reality Technologies like MR which stands for Mixed Reality and XR as EXtended Reality which are in modern World

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Question: Are AR and VR Booming the Industry?

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Question: Are AR and VR Booming the Industry?

!NO!

Allowing computer-generated items to interact with tangible objects blurs the line between real and virtual interaction

In Addition to AR and VR there Exist other Reality Technologies like **MR which stands for Mixed Reality** and XR as EXtended Reality which are in modern World

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Question: Are AR and VR Booming the Industry?

!NO!

It combines all physical and virtual environments created by computer graphics and wearables

In Addition to AR and VR there Exist [illegible] Reality Technologies like MR which stands for Mixed Reality and **XR as EXtended Reality** which are in modern World

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Companies Using and
Hiring AR VR
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Companies Using and Hiring VR and AR

Google

KARUNA

BMW

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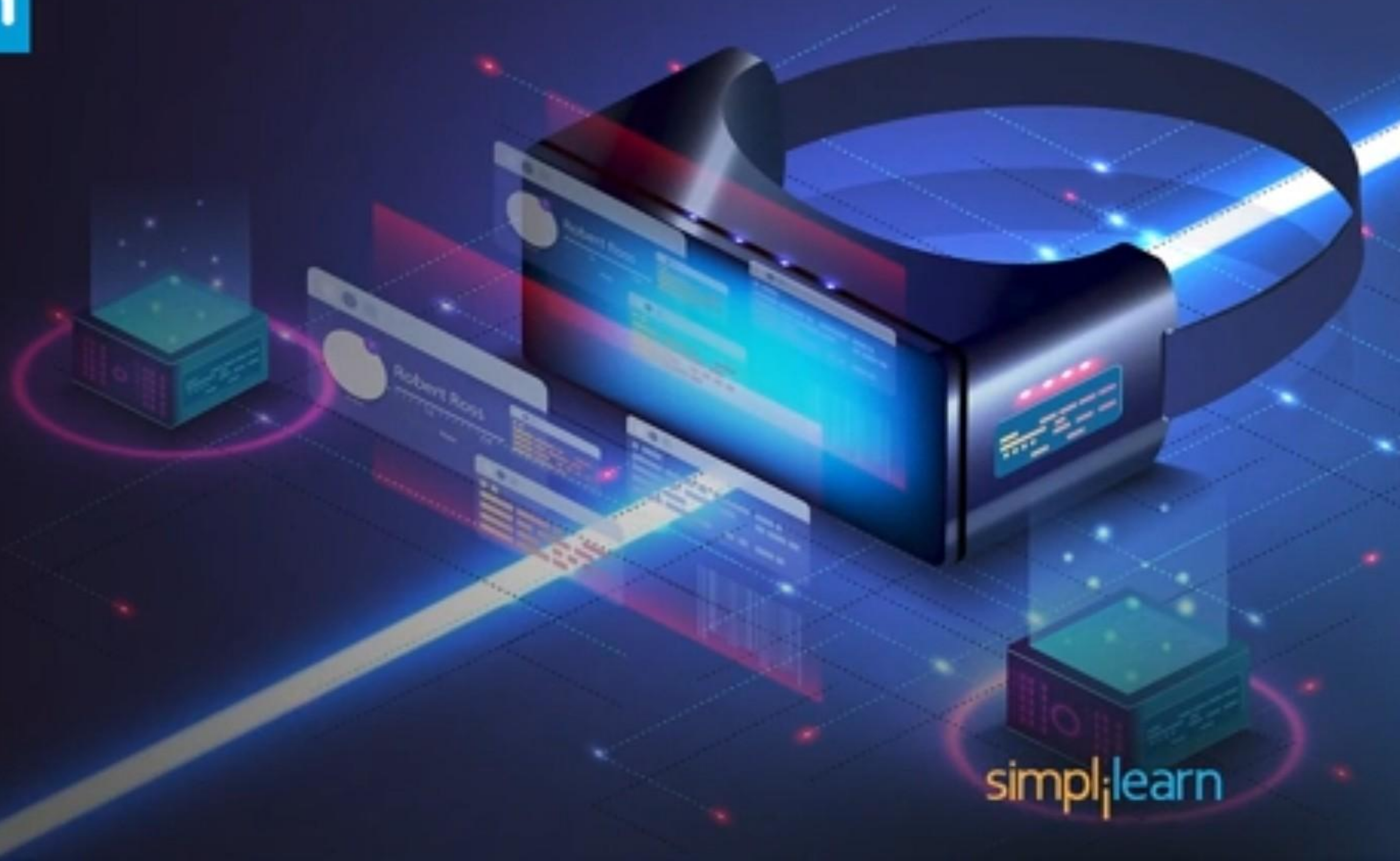
Conclusion



Conclusion

In this module we have discussed about

- VR and AR
- Its applications and real world applications
- Advantages and disadvantages of VR and AR
- And Its implementation and its real world usage.

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Agenda

01 What is AR?
02 Need For AR
03 Best AR Apps
04 Comparison
05 Conclusion

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What is AR?

Augmented reality (AR) adds digital elements to a live view often by using the camera on a smartphone

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For Example

Google Maps

SCOPE AR

ARCore
by Google

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Need For AR

- AR covers digital information on physical one
- It helps produce remote operations
- AR enhances better Decision Making
- It Transform data and analytics into images or animations
- AR is easy Accessible

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Best Apps in 2022

1.Wikitude

Wikitude is one of the extensively used AR platform to develop AR mobile apps and prototypes. It allows developers to implement geolocation features, along with image tracking and object recognition

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2.Apple AR kit

This AR platform has made it easier for developers to design and develop applications, proceed with facial tracking, stable and fast motion tracking, etc

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3.Augmentir

Augmentir is the world's only AI-Powered Connected Worker platform. playing a vital role in helping out manufacturing and service companies to improve the safety, quality, and productivity

Augmentir

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4.ARCore

ARCore is one among other AR platform that follows three key technologies for 'embedding' virtual content into real environments namely motion tracking, environmental recognition, and lighting recognition

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5.Vuforia

Vuforia, a powerful AR platform, which is aimed at developing AR applications with the objective of enabling businesses to provide an immersive AR experience to their customers

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Comparison Time

Key Factors of Comparison

Factors
1.Cost Effective
2.Security
3.AI
4.Multi-Platform
5.Storage

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Comparison Time

Factors	Cost Effective	Security	AI	Multi platform	Storage
Wikitude					
Apple AR kit					
Augmentir					
ARCore					
Vuforia					

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Comparison Time

Factors	Cost Effective	Security	AI	Multi platform	Storage
Wikitude	NO	NA	NO	YES	YES
Apple AR kit					
Augmentir					
ARCore					
Vuforia					

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Comparison Time

Factors	Cost Effective	Security	AI	Multi platform	Storage
Wikitude	NO	NA	NO	YES	YES
Apple AR kit	NO	YES	NA	NO	NO
Augmentir					
ARCore					
Vuforia					

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Comparison Time

Factors	Cost Effective	Security	AI	Multi platform	Storage
Wikitude	NO	NA	NO	YES	YES
Apple AR kit	NO	YES	NA	NO	NO
Augmentir	NA	YES	YES	YES	NO
ARCore					
Vuforia					

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Comparison Time

Factors	Cost Effective	Security	AI	Multi platform	Storage
Wikitude	NO	NA	NO	YES	YES
Apple AR kit	NO	YES	NA	NO	NO
Augmentir	NA	YES	YES	YES	NO
ARCore	YES	YES	YES	YES	YES
Vuforia					

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parison Time

tors	Cost Effective	Security	AI	Multi platform	Storage
tude	NO	NA	NO	YES	YES
AR kit	NO	YES	NA	NO	NO
nentir	NA	YES	YES	YES	NO
Core	YES	YES	YES	YES	YES
oria	NO	YES	YES	YES	YES

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Conclusion

This module covers

- What is AR
- Need For AR
- Top AR apps in 2022
- Its key features and
- Comparison on key factors

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